

ASSIGNMENT

- 1 If the roots of the equation $x^3 - 12x^2 + 39x - 28 = 0$ are in A.P., then their common difference will be

Option:

(a) ± 1

(b) ± 2

(c) ± 3

(d) ± 4

Answer: (c)

Solution:

Let $a - d, a, a + d$ be the roots of the equation

$$x^3 - 12x^2 + 39x - 28 = 0$$

$$\text{Then } (a - d) + a + (a + d) = 12 \text{ and } (a - d)a(a + d) = 28$$

$$\Rightarrow 3a = 12 \text{ and } a(a^2 - d^2) = 28$$

$$\Rightarrow a = 4 \text{ and } a(a^2 - d^2) = 28$$

$$\Rightarrow 16 - d^2 = 7 \Rightarrow d = \pm 3$$

- 2 The sixth term of an A.P. is equal to 2. Find the value of the common difference of the A.P. which makes the product $a_4 a_5 a_7 a_8$ least.

Solution:

Let a be the first term and x be the common difference of the A.P. Then $a + 5x = 2 \Rightarrow a = 2 - 5x$

$$\text{Let } P = a_4 a_5 a_7 a_8 = (a - 2x)(a - x)(a + x)(a + 2x)$$

$$= (2 - 2x)(2 - x)(2 + x)(2 + 2x)$$

$$= 4(4 - x^2)(1 - x^2)$$

$$= 4(x^4 - 5x^2 + 4)$$

$$= 4\left(x^2 - \frac{5}{2}\right)^2 - 9$$

$$\text{Hence } P \text{ is least for } x = \pm \frac{\sqrt{5}}{2}$$

- 3 If p, q, r are in A.P. and are positive, the roots of the quadratic equation $px^2 + qx + r = 0$ are all real for

Option:

(a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$

(b) $\left|\frac{p}{r} - 7\right| < 4\sqrt{3}$

(c) All p and r

(d) No p and r

Answer: (a)

Solution:

p, q, r are positive and are in A.P.

$$\therefore q = \frac{p+r}{2} \quad \dots\dots(i)$$

$\therefore$ The roots of $px^2 + qx + r = 0$ are real

$$\Rightarrow q^2 \geq 4pr \Rightarrow \left[\frac{p+r}{2} \right]^2 \geq 4pr \quad [\text{using (i)}]$$

$$\Rightarrow p^2 + r^2 - 4pr \geq 0$$

$$\Rightarrow \left(\frac{r}{p} \right)^2 - 4 \left(\frac{r}{p} \right) + 1 \geq 0 \quad (\because p > 0 \text{ and } p \neq 0)$$

$$\Rightarrow \left(\frac{r}{p} - 2 \right)^2 - 3 \geq 0 \Rightarrow \left(\frac{r}{p} - 2 \right)^2 - (\sqrt{3})^2 \geq 0$$

$$\Rightarrow \left| \frac{r}{p} - 2 \right| \geq \sqrt{3}$$

- 4 If the sum of three numbers of an arithmetic sequence is 15 and the sum of their squares is 83, find the numbers.

Solution:

Let three numbers be $a-d, a, a+d$.

$$\text{We get } a-d+a+a+d=15 \Rightarrow a=5$$

$$\text{and } (a-d)^2 + a^2 + (a+d)^2 = 83$$

$$\Rightarrow a^2 + d^2 - 2ad + a^2 + a^2 + d^2 + 2ad = 83$$

$$\Rightarrow 2(a^2 + d^2) + a^2 = 83$$

$$\text{Putting } a=5$$

$$\Rightarrow 2(25 + d^2) + 25 = 83 \Rightarrow 2d^2 = 8 \Rightarrow d = 2$$

Thus numbers are 3, 5, 7.

- 5 Four numbers are in arithmetic progression. The sum of first and last term is 8 and the product of both middle terms is 15. Find the least number of the series.

Solution:

Let A_1, A_2, A_3 and A_4 are four numbers in A.P.

$$A_1 + A_4 = 8 \quad \dots\dots(i) \quad \text{and } A_2 \cdot A_3 = 15 \quad \dots\dots(ii)$$

The sum of terms equidistant from the beginning and end is constant and is equal to sum of first and last terms.

$$\text{Hence, } A_2 + A_3 = A_1 + A_4 = 8 \quad \dots\dots(iii)$$

From (ii) and (iii),

$$A_2 + \frac{15}{A_2} = 8 \Rightarrow A_2^2 - 8A_2 + 15 = 0$$

$A_2 = 3$ or 5 and $A_3 = 5$ or 3 .

As we know, $A_2 = \frac{A_1 + A_3}{2} \Rightarrow A_1 = 2A_2 - A_3$

$\Rightarrow A_1 = 2 \times 3 - 5 = 1$ and $A_4 = 8 - A_1 = 7$

Hence the series is, 1, 3, 5, 7.

So that least number of series is 1.

- 6 If S_n denotes the sum of n terms of an arithmetic progression, then the value of $(S_{2n} - S_n)$ is equal to

Option:

- (a) $2S_n$ (b) S_{3n} (c) $\frac{1}{3}S_{3n}$ (d) $\frac{1}{2}S_n$

Answer: (c)

Solution:

$$\begin{aligned} S_{2n} - S_n &= \frac{2n}{2} \{2a + (2n-1)d\} - \frac{n}{2} \{2a + (n-1)d\} \\ &= \frac{n}{2} \{4a + 4nd - 2d - 2a - nd + d\} = \frac{n}{2} \{2a + (3n-1)d\} \\ &= \frac{1}{3} \cdot \frac{3n}{2} \{2a + (3n-1)d\} = \frac{1}{3} S_{3n} \end{aligned}$$

- 7 If $a_1, a_2, \dots, a_n$ are in A.P. with common difference, d , then the sum of the following series is $\sin d(\operatorname{cosec} a_1 \cdot \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \cdot \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n)$

Option:

- (a) $\sec a_1 - \sec a_n$ (b) $\cot a_1 - \cot a_n$ (c) $\tan a_1 - \tan a_n$
(d) $\operatorname{cosec} a_1 - \operatorname{cosec} a_n$

Answer: (b)

Solution:

As given $d = a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1}$

$\therefore \sin d \{ \operatorname{cosec} a_1 \operatorname{cosec} a_2 + \dots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n \}$

$$\begin{aligned} &= \frac{\sin(a_2 - a_1)}{\sin a_1 \cdot \sin a_2} + \dots + \frac{\sin(a_n - a_{n-1})}{\sin a_{n-1} \sin a_n} \\ &= (\cot a_1 - \cot a_2) + (\cot a_2 - \cot a_3) + \dots \\ &\quad + (\cot a_{n-1} - \cot a_n) \\ &= \cot a_1 - \cot a_n. \end{aligned}$$

- 8 Let $\{a_n\}$ ($n \geq 1$) be a sequence such that $a_1 = 1$, and $3a_{n+1} - 3a_n = 1$ for all $n \geq 1$. Then a_{2002} is equal to-

Option:

- (a) 666 (b) 667 (c) 668 (d) 669

Answer: (c)

Solution:

$$a_1 = 1$$

$$3a_{n+1} - 3a_n = 1$$

$$a_{n+1} = \frac{3a_n + 1}{3} = a_n + \frac{1}{3},$$

$$a_2 = a_1 + \frac{1}{3} = 1 + \frac{1}{3}$$

$$a_3 = a_2 + \frac{1}{3} = a_1 + \frac{1}{3} + \frac{1}{3} = 1 + \frac{2}{3}$$

$$a_4 = a_3 + \frac{1}{3} = 1 + \frac{2}{3} + \frac{1}{3} = 1 + \frac{3}{3}$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$a_{2002} = 1 + \frac{2001}{3} = 1 + 667 = 668$$

- 9 If the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$ are in G.P., then

Option:

- (a) $c^3a = b^3d$ (b) $ca^3 = bd^3$ (c) $a^3b = c^3d$ (d) $ab^3 = cd^3$

Answer: (a)

Solution:

Let $\frac{A}{R}$, A, AR be the roots of the equation

$$ax^3 + bx^2 + cx + d = 0$$

$$\text{then } A^3 = \text{Product of the roots} = -\frac{d}{a} \Rightarrow A = -\left(\frac{d}{a}\right)^{1/3}$$

Since A is a root of the equation.

$$\therefore aA^3 + bA^2 + cA + d = 0$$

$$\Rightarrow a\left(-\frac{d}{a}\right) + b\left(-\frac{d}{a}\right)^{2/3} + c\left(-\frac{d}{a}\right)^{1/3} + d = 0 \Rightarrow b\left(\frac{d}{a}\right)^{2/3} = c\left(\frac{d}{a}\right)^{1/3} \Rightarrow b^3\frac{d^2}{a^2} = c^3\frac{d}{a} \Rightarrow b^3d = c^3a.$$

- 10 If the first term of a G.P. $a_1, a_2, a_3, \dots$ is unity such that $4a_2 + 5a_3$ is least, then the common ratio of G.P. is

Option:

- (a) $-\frac{2}{5}$ (b) $-\frac{3}{5}$ (c) $\frac{2}{5}$ (d) None of these

Answer: (a)

Solution:

$$a_1 = 1, a_2 = r, a_3 = r^2, \dots$$

$$\therefore 4a_2 + 5a_3 = 4r + 5r^2 = 5\left(r + \frac{2}{5}\right)^2 - \frac{4}{5}$$

To be its minimum $r = \frac{-2}{5}$.

- 11 The least positive integer n such that $1 - \frac{2}{3} - \frac{2}{3^2} - \dots - \frac{2}{3^{n-1}} < \frac{1}{100}$ is

Solution:

Let S_n denote the sum to n terms of the given series

$$S_n = 1 - \frac{2}{3} - \frac{2}{3^2} - \frac{2}{3^3} \dots - \frac{2}{3^{n-1}}$$

$$= 1 - \frac{\left(\frac{2}{3}\right)\left[1 - \left(\frac{1}{3}\right)^{n-1}\right]}{1 - \frac{1}{3}}$$

$$= 1 - 1 + \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{n-1}$$

$$\text{Now, } \left(\frac{1}{3}\right)^{n-1} < \frac{1}{100}$$

$$\Rightarrow 3^{n-1} > 100 \Rightarrow n-1 \geq 5 \Rightarrow n \geq 6$$

Thus, least value of n is 6.

- 12 The sum of a G.P. with common ratio 3 is 364, and last term is 243, then the number of terms is

Option:

- (a) 6 (b) 5 (c) 4 (d) 10

Answer: (a)

Solution:

$$\text{Given } S_n = 364, r = 3, T_n = 243$$

$$\frac{ar^n - a}{r - 1} = 364 \Rightarrow \frac{ar^{n-1} \cdot r - a}{r - 1} = 364 \dots (1)$$

$$\Rightarrow \frac{3 \times 243 - a}{2} = 364 \Rightarrow a = 1$$

Now, putting this in (1)

$$\frac{r^n - 1}{r - 1} = 364$$

$$\frac{3^n - 1}{2} = 364$$

$$3^n - 1 = 728$$

$$3^n = 729$$

$$3^n = 3^6$$

$$\therefore n = 6$$

- 13 The sum of n terms of the following series $1 + (1 + x) + (1 + x + x^2) + \dots$ will be

Option:

- (a) $\frac{1 - x^n}{1 - x}$ (b) $\frac{x(1 - x^n)}{1 - x}$ (c) $\frac{n(1 - x) - x(1 - x^n)}{(1 - x)^2}$ (d) None of these

Answer: (c)

Solution:

$$1 + (1 + x) + (1 + x + x^2) + \dots +$$

$$(1 + x + x^2 + x^3 + \dots + x^{n-1}) + \dots$$

$$\text{Required sum} = \frac{1}{(1 - x)} \{ (1 - x) + (1 - x^2) + (1 - x^3) + (1 - x^4) + \dots \text{upto } n \text{ terms} \}$$

$$= \frac{1}{(1 - x)} [n - \{x + x^2 + x^3 + \dots \text{upto } n \text{ terms}\}]$$

$$= \frac{1}{(1 - x)} \left[n - \frac{x(1 - x^n)}{1 - x} \right] = \frac{n(1 - x) - x(1 - x^n)}{(1 - x)^2}$$

- 14 If the sum of first 6 term is 9 times to the sum of first 3 terms of the same G.P., then the common ratio of the series will be

Option:

- (a) -2 (b) 2 (c) 1 (d) $\frac{1}{2}$

Answer: (b)

Solution:

Given conditions, we get

$$\Rightarrow \frac{a(r^6-1)}{(r-1)} = 9 \cdot \frac{a(r^3-1)}{(r-1)}$$

$$\Rightarrow (r^6-1) = 9(r^3-1)$$

$$(r^3+1)(r^3-1) = 9(r^3-1)$$

$$r^3+1=9$$

$$r^3=8$$

$$\therefore r=2$$